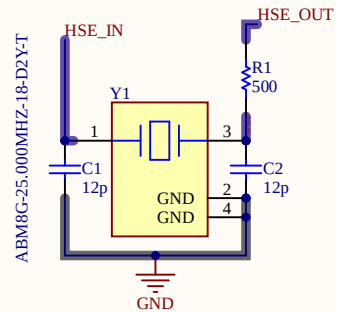
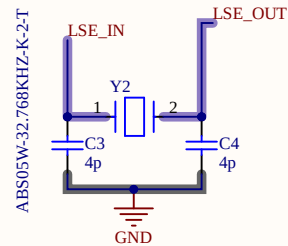


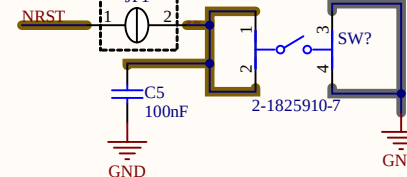


High Speed Oscillator 25MHz

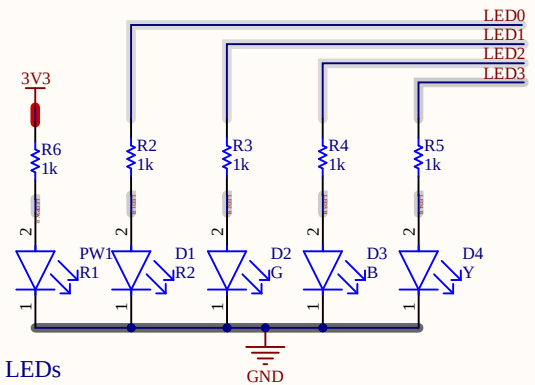


Low Speed Oscillator 32KHz

2 Pin jumper to disable the reset button line if required.

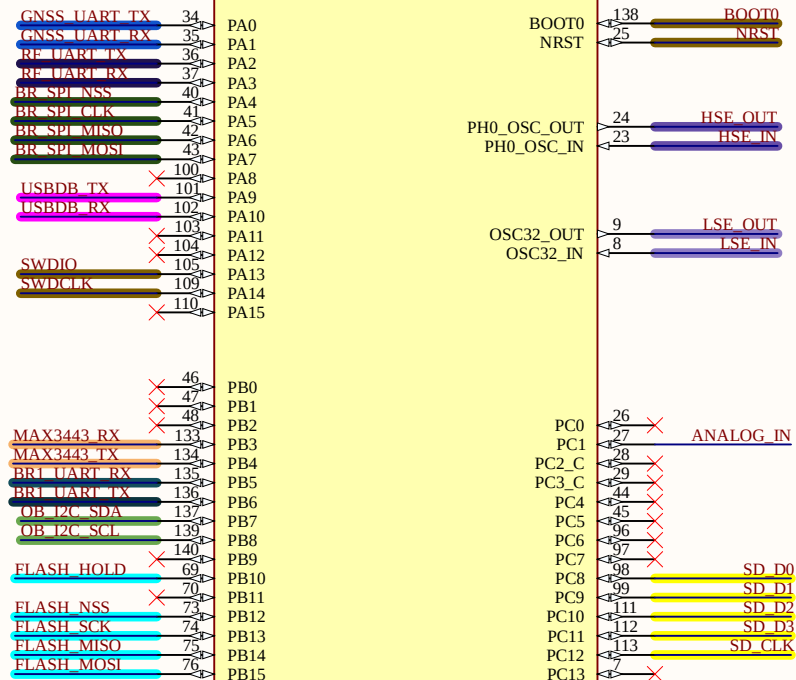


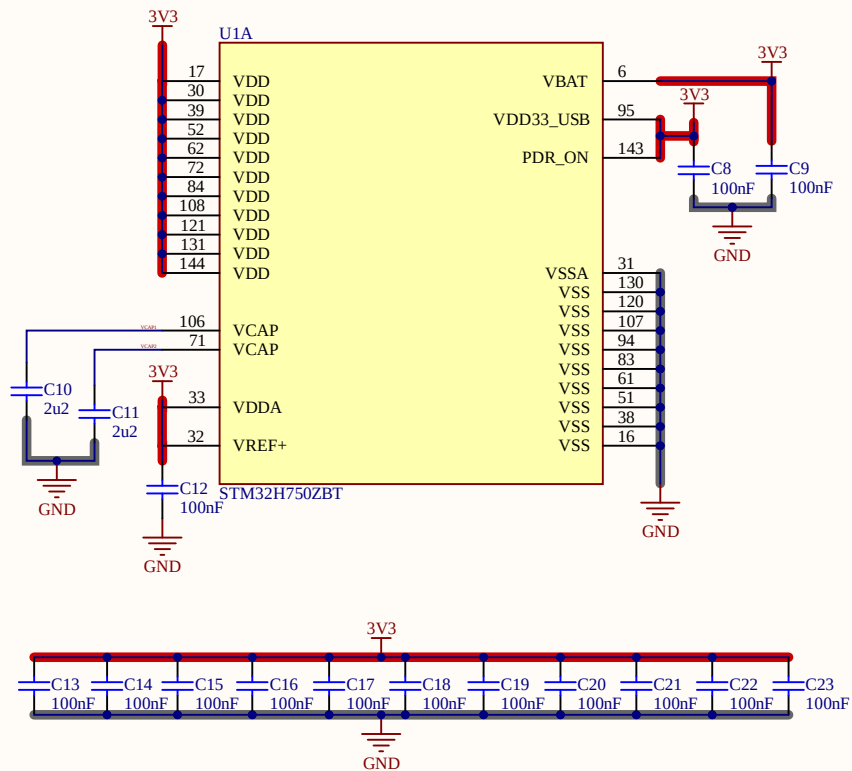
Reset Button



Prog LEDs

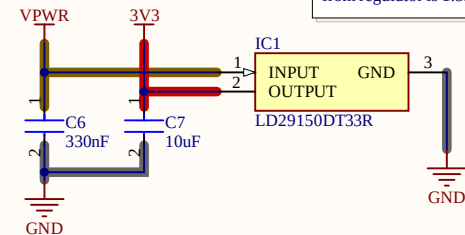
U1B





MCU Power

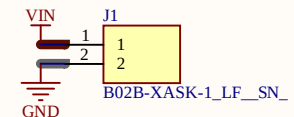
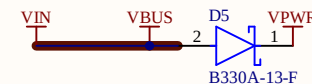
3V3 Regulator



△ Takes input from the 5V/VBUS rail and converts to 3V3 for all components.

Max regulator input is 30V and Max current throughput from regulator is 1.5A at 3.3V.

△ Power port and USB bus are connected to the same rail. Board can be driven through both USB and power port



△ Isolated 5V output from the EPS

Power Port

A

A

B

B

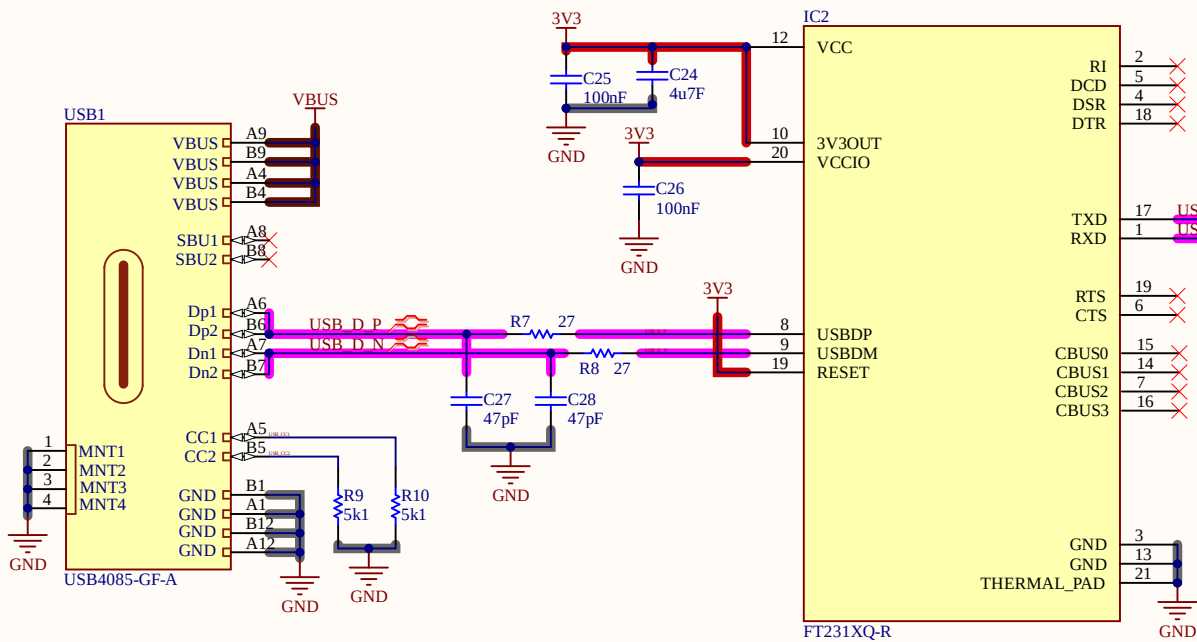
C

C

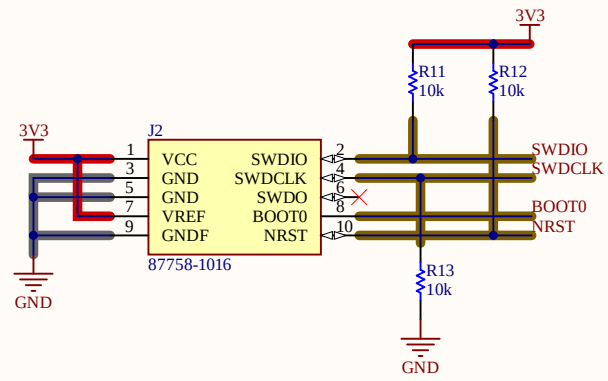
D

D

USB to UART Debug

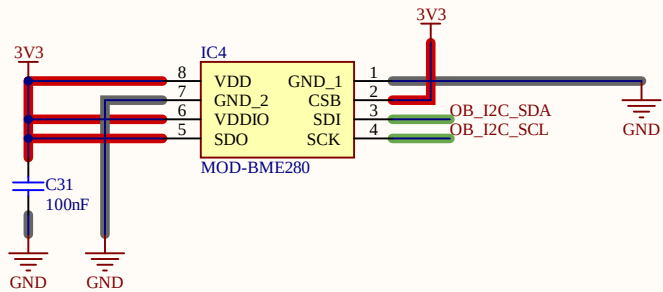


SWD 2x5 2mm Pin Header



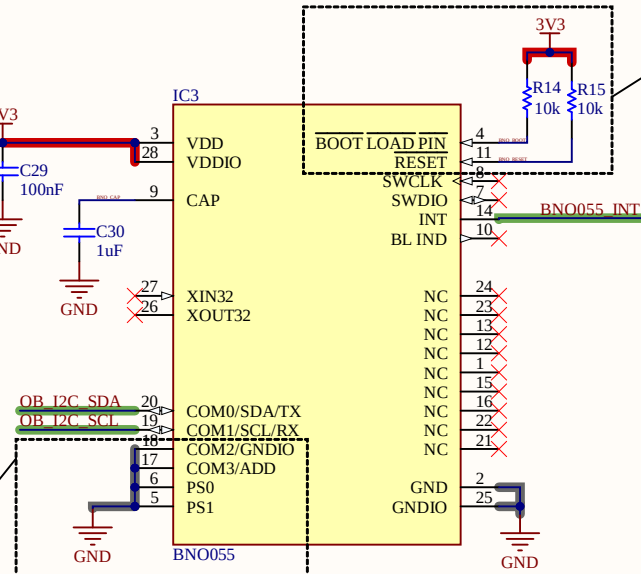
Title		
Size	Number	Revision
A4		
Date:	1/11/2026	Sheet of
File:	Programming_Interface.SchDoc	Drawn By:

A



BME280
Temperature & Humidity Sensor

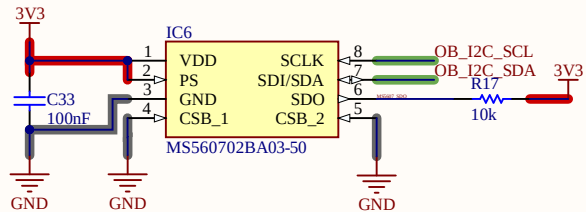
B



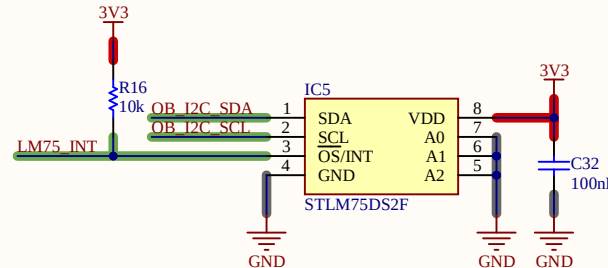
BNO055
9-Axis IMU

COM3 acts as address pin when operating in I2C mode. Address has been set to 0x28 by pulling it low (0x29 at high).
PS0/PS1 are protocol select pins. Both pulled low to configure the sensor to I2C interface mode

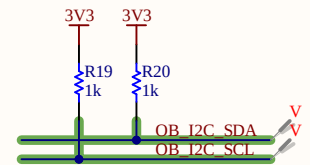
C



MS56070
Pressure Sensor



STLM75
Temperature Sensor

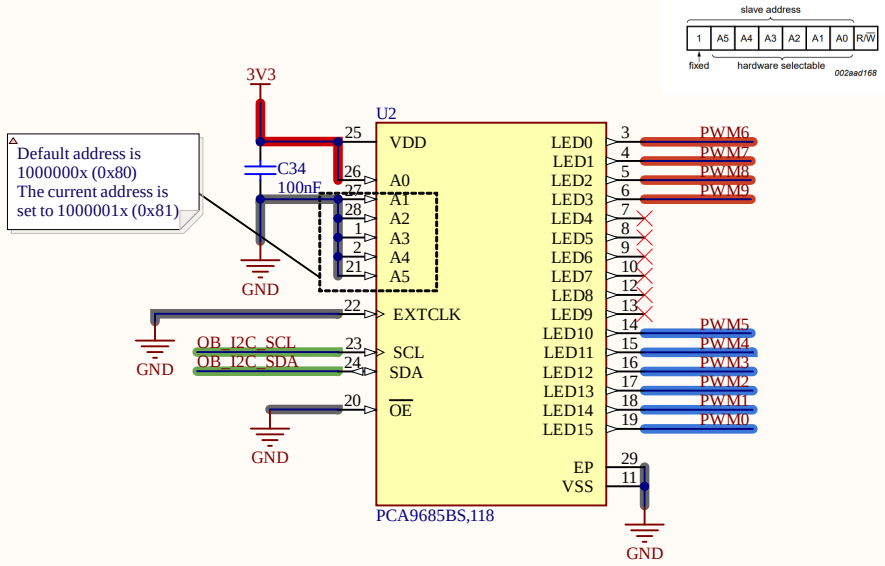


Title		
Size	Number	Revision
A4		
Date:	1/11/2026	Sheet of
File:	AD_Sensors.SchDoc	Drawn By:

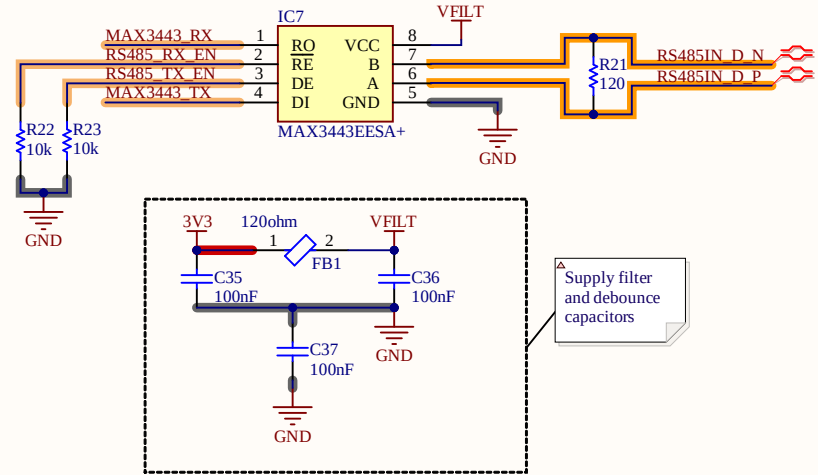
D

D

A

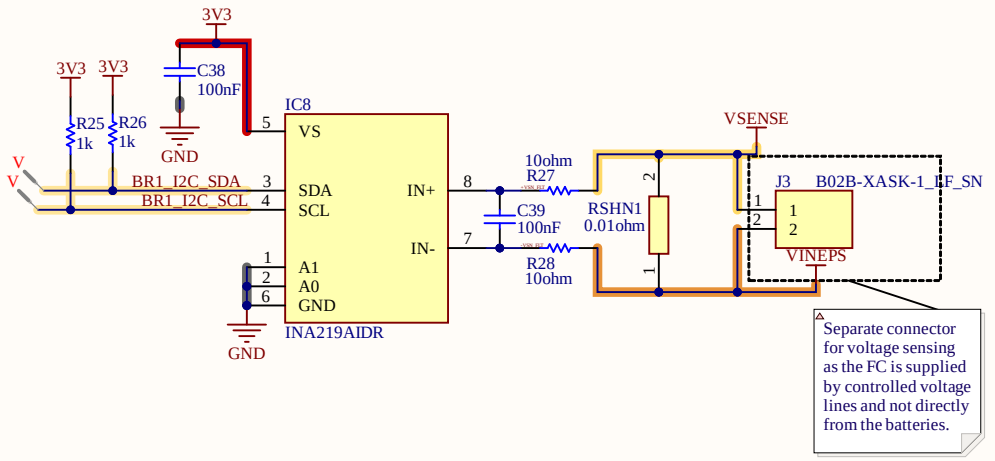


PCA9685

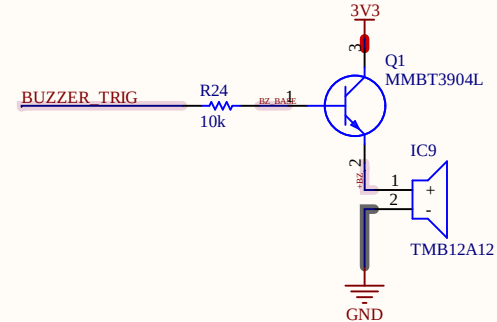


Breakout RS485

C



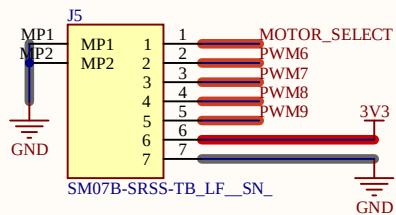
INA219



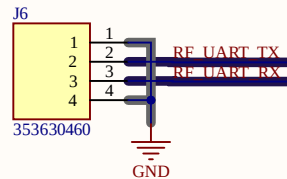
Passive Buzzer

D

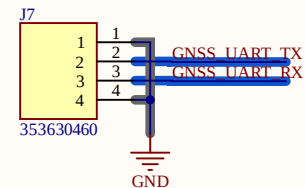
Title		
Size	Number	Revision
A4		
Date:	1/11/2026	Sheet of
File:	Support_Peripherals.SchDoc	Drawn By:



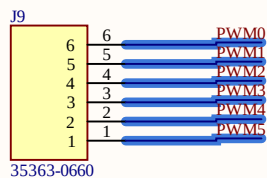
ESC Connector (1x7)



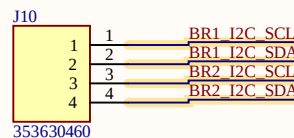
RF PCB Connect (1x4)



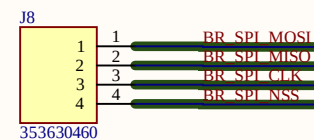
GNSS UART (1x4)



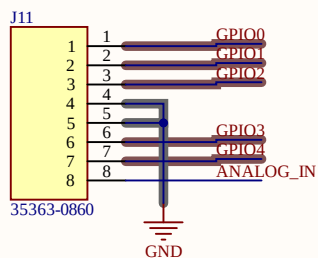
Servo Control (1x6)



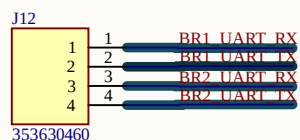
Breakout I2C (1x4)



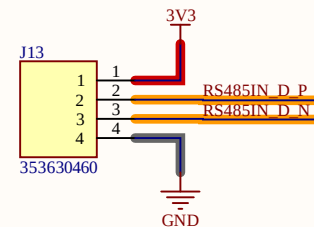
Breakout SPI (1x4)



Breakout GPIO (1x8)



Breakout UART (1x4)



Title		
Size	Number	Revision
Orcad A		
Date:	1/11/2026	Sheet of
File:	Connectors.SchDoc	Drawn By: